



Comprehensive Compression Benchmark Results

Project: TAI - Algorithmic Information Theory

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Tools Compared: G07, gzip, bzip2, xz (LZMA), zstd

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Quick Summary

Compression Ratio Comparison

File	Size	G07	gzip	bzip2	xz	zstd	Winner
A	1.3MiB	53.39% ★	58.49%	54.05%	53.34% ★	53.52%	xz
B	1.2MiB	18.81% ★	22.89%	17.92% ★	17.76% ★	23.11%	xz
C	2.0MiB	27.30% ★	32.57%	26.51% ★	24.72% ★	31.19%	xz
D	2.0MiB	100.00%	100.02%	100.45%	100.01%	100.00%	g07

File	Size	G07	gzip	bzip2	xz	zstd	Winner
		★					
E	989KiB	88.41% ★	88.66%	88.43%	85.84% ★	88.53%	xz
F	2.0MiB	88.04% ★	88.01% ★	81.06% ★	82.38%	87.81%	bzip2
G	2.5MiB	29.12% ★	36.99%	28.70% ★	29.71%	35.82%	bzip2
H	999KiB	45.54% ★	44.02% ★	42.34% ★	35.88% ★	44.86%	xz

★ = Best compressor for that file

Detailed Results by File

File: A

Original Size: 1308765 bytes (1.3MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	683KiB (698813)	 53.39%	4.2716	113 ms	71	✓ 596KiB saved	✓
gzip	748KiB (765497)	 58.49%	4.6792	55 ms	16	✓ 531KiB saved	✓
bzip2	691KiB (707422)	 54.05%	4.3242	105 ms	63	✓ 588KiB	✓

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
						saved	
xz	682KiB (698048)	53.34%	4.2669	359 ms	35	✓ 597KiB saved	✓
zstd	684KiB (700409)	53.52%	4.2813	15 ms	12	✓ 595KiB saved	✓

File: B

Original Size: 1168767 bytes (1.2MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	215KiB (219821)	18.81%	1.5046	65 ms	51	✓ 927KiB saved	✓
gzip	262KiB (267569)	22.89%	1.8315	48 ms	11	✓ 881KiB saved	✓
bzip2	205KiB (209479)	17.92%	1.4338	90 ms	35	✓ 937KiB saved	✓
xz	203KiB (207584)	17.76%	1.4209	269 ms	19	✓ 939KiB saved	✓
zstd	264KiB (270069)	23.11%	1.8486	23 ms	12	✓ 878KiB saved	✓

File: C

Original Size: 2000000 bytes (2.0MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	534KiB (545915)	27.30%	2.1837	99 ms	74	✓ 1.4MiB saved	✓
gzip	637KiB (651482)	32.57%	2.6059	65 ms	16	✓ 1.3MiB saved	✓
bzip2	518KiB (530253)	26.51%	2.1210	126 ms	62	✓ 1.5MiB saved	✓
xz	483KiB (494424)	24.72%	1.9777	508 ms	32	✓ 1.5MiB saved	✓
zstd	610KiB (623877)	31.19%	2.4955	24 ms	17	✓ 1.4MiB saved	✓

File: D

Original Size: 2000000 bytes (2.0MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	2.0MiB (2000009)	100.00%	8.0000	19 ms	16	◇ overhead	✓
gzip	2.0MiB (2000330)	100.02%	8.0013	73 ms	9	◇ overhead	✓

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
bzip2	2.0MiB (2009059)	100.45%	8.0362	183 ms	113	? overhead	✓
xz	2.0MiB (2000168)	100.01%	8.0007	385 ms	11	? overhead	✓
zstd	2.0MiB (2000061)	100.00%	8.0002	17 ms	15	? overhead	✓

File: E

Original Size: 1012112 bytes (989KiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	874KiB (894847)	88.41%	7.0731	30 ms	61	✓ 115KiB saved	✓
gzip	877KiB (897332)	88.66%	7.0927	47 ms	12	✓ 113KiB saved	✓
bzip2	875KiB (895061)	88.43%	7.0748	105 ms	62	✓ 115KiB saved	✓
xz	849KiB (868764)	85.84%	6.8669	197 ms	37	✓ 140KiB saved	✓
zstd	875KiB (895975)	88.53%	7.0820	9 ms	16	✓ 114KiB saved	✓

File: F

Original Size: 2097152 bytes (2.0MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	1.8MiB (1846298)	88.04%	7.0431	39 ms	106	✓ 245KiB saved	✓
gzip	1.8MiB (1845625)	88.01%	7.0405	74 ms	20	✓ 246KiB saved	✓
bzip2	1.7MiB (1699879)	81.06%	6.4845	170 ms	112	✓ 388KiB saved	✓
xz	1.7MiB (1727576)	82.38%	6.5902	458 ms	62	✓ 361KiB saved	✓
zstd	1.8MiB (1841432)	87.81%	7.0245	12 ms	17	✓ 250KiB saved	✓

File: G

Original Size: 2526752 bytes (2.5MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	719KiB (735697)	29.12%	2.3293	129 ms	84	✓ 1.8MiB saved	✓
gzip	913KiB		2.9592	239 ms	15	✓	✓

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
	(934659)	36.99%				1.6MiB saved	
bzip2	709KiB (725142)	28.70%	2.2959	173 ms	86	✓ 1.8MiB saved	✓
xz	734KiB (750608)	29.71%	2.3765	1025 ms	35	✓ 1.7MiB saved	✓
zstd	884KiB (905141)	35.82%	2.8658	26 ms	20	✓ 1.6MiB saved	✓

File: H

Original Size: 1022760 bytes (999KiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	455KiB (465789)	45.54%	3.6434	116 ms	102	✓ 544KiB saved	✓
gzip	440KiB (450169)	44.02%	3.5212	52 ms	13	✓ 560KiB saved	✓
bzip2	423KiB (432998)	42.34%	3.3869	82 ms	46	✓ 576KiB saved	✓
xz	359KiB (366916)	35.88%	2.8700	231 ms	26	✓ 641KiB	✓

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
						saved	
zstd	449KiB (458829)	44.86%	3.5889	21 ms	16	✓ 551KiB saved	✓

Performance Metrics

Compression Speed (Average)

Tool	Avg Compress Time	Ranking
zstd	18 ms	 #1
g07	76 ms	 #2
gzip	81 ms	 #3
bzip2	129 ms	#4
xz	429 ms	#5

Decompression Speed (Average)

Tool	Avg Decompress Time	Ranking
gzip	14 ms	 #1
zstd	15 ms	 #2
xz	32 ms	 #3
g07	70 ms	#4
bzip2	72 ms	#5

Rankings

Overall Compression Ratio Ranking

Rank	Tool	Average Ratio	Files Won	Total Compressed	Performance
 1	xz	54.16%	5	6.8MiB	★ ★ ★
 2	bzip2	54.88%	2	6.9MiB	★ ★
 3	g07	56.39%	1	7.1MiB	★
4	zstd	58.58%	0	7.4MiB	
5	gzip	59.47%	0	7.5MiB	

Files Won by Each Tool

- **g07**: 1 file(s)
- **gzip**: 0 file(s)
- **bzip2**: 2 file(s)
- **xz**: 5 file(s)
- **zstd**: 0 file(s)

Overall Statistics

Tool	Total Original	Total Compressed	Average Ratio	Bits/ Symbol	Files Tested
g07	13MiB	7.1MiB	56.39%	4.5110	8
gzip	13MiB	7.5MiB	59.47%	4.7579	8
bzip2	13MiB	6.9MiB	54.88%	4.3905	8

Tool	Total Original	Total Compressed	Average Ratio	Bits/ Symbol	Files Tested
xz	13MiB	6.8MiB	54.16%	4.3325	8
zstd	13MiB	7.4MiB	58.58%	4.6867	8



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